

1. Name of the medicinal product

DIACOL, 27 mg/15 ml, syrup

2. Qualitative and quantitative composition

27 mg dextromethorphan hydrobromide per 15 ml of syrup.

For excipients, see 6.1.

3. Pharmaceutical form

Syrup.

4. Clinical particulars**4.1. Therapeutic indications**

Temporary relief of non-productive cough due to irritation of upper and lower respiratory tract.

4.2. Posology and administration

Unless your doctor advises otherwise, DIACOL should be taken as follows:

Adults, elderly and adolescents: 15 ml every 6 or 8 hours. Maximum daily dose is 60 ml.

Children 2 to 12 years: dosage according to medical prescription. *Children aged 6 to 12 years old*, maximum daily dose is 30 ml. *Children aged 2 to 6 years old*, maximum daily dose is 15 ml.

Children under 2 years: Not recommended for this age group.

DIACOL may be administered every 4 hours, reducing the amount of syrup per intake, not exceeding the maximum daily dose. Do not exceed the recommended doses or use for long periods. Take it alone or diluted in a small amount of water.

4.3. Contraindications

Hypersensitivity to any of the ingredients. Productive cough. Concomitant treatment with monoamine oxidase inhibitors (MAO).

4.4. Special warnings and special precautions for use

DIACOL should be used with caution in asthmatic or atopic patients, since it may promote histamine release. And also in sedated or debilitated patients.

In patients with hepatic impairment it may occur metabolism changes.

If cough persists for longer than 1 week, tends to recur, or is accompanied by high fever, skin eruption or strong headache, a physician should be consulted.

DIACOL contains 121.64 g sucrose per 200 ml of syrup. A 15 ml-dose contains approximately 9.1 g sucrose. Do not use in case of inherited fructose intolerance, glucose-galactose malabsorption syndrome or sucrase / isomaltase deficiency.

DIACOL contains 2.75 g of ethanol 96% per 200 ml of syrup. A 15 ml-dose contains approximately 0.20 g of ethanol 96%. It can be harmful for those suffering from liver disease, alcoholism, epilepsy, brain injury or disease, as well as for pregnant women and children. May modify or increase the effect of other medicines.

4.5. Interactions with other medicinal products and other forms of interaction

Concurrent use of DIACOL with monoamine (MAO) oxidase inhibitors may promote adrenergic crisis, coma, vertigo, excitability, hypertension, hyperpyrexia, intracerebral haemorrhage, lethargy, nausea, psychotic behaviours, muscle spasms, tremors and, eventually death and therefore concomitant administration is contraindicated.

An interval of 14 days should be respected between withdrawal of monoamine oxidase inhibitors and DIACOL intake.

Effects of medicines acting upon the central nervous system may be enhanced when used concurrently with dextromethorphan, and therefore it should be taken into account.

There may be an increased risk of adverse reactions if DIACOL is taken concomitantly with quinidine, fluoxetine, paroxetine and venlafaxine

Alcoholic drinks or medications containing alcohol are not recommended while taking concurrently DIACOL.

4.6. Pregnancy and lactation

Although there are no reports of foetal problems secondary to the use of DIACOL during pregnancy or interference on the reproduction system DIACOL is not recommended during pregnancy. Also in lactation.

4.7. Effects on ability to drive and use machines

No known interferences at the recommended therapeutic doses.

4.8. Undesirable effects

Usually, side effects due to DIACOL are rare, and may appear as slight gastrointestinal disturbances, namely nausea and vomiting, stomach pains, drowsiness and vertigo.

4.9. Overdose

Manifestations following acute overdosage have included nausea, vomiting, drowsiness, blurred vision, nystagmus, ataxia, urinary retention, shallow respiration depression, stupor, toxic psychosis (hyperactivity, visual and auditory hallucinations) and coma. Treatment of dextromethorphan overdosage includes symptomatic and supportive measures and parenteral administration of naloxone and oral administration of activated charcoal.

5. Pharmacological properties**5.1. Pharmacodynamic properties**

ATC Classification: R05D A09.

Dextromethorphan is a central acting antitussive. It is an isomer of levorphanol, a codeine analogue. Unlike the other isomers, dextromethorphan has no analgesic or addictive properties, and does not act by means of opiate receptors. Its antitussive properties result from its direct action on the cough centre in the medulla, raising the threshold for the cough reflex. Its antitussive strength is similar to codeine, persisting its effects for 6 hours. Contrarily to codeine, its sedative action is practically negligible, since it only occurs at extremely high doses.

In therapeutic dosage, it does not inhibit the ciliary function of airways, thus providing a clinically advantage over many other cough suppressants.

5.2. Pharmacokinetic properties

Dextromethorphan is rapidly absorbed from the gastrointestinal tract. T_{max} is 2 to 2.5 hours and C_{max} after an oral dose of 60 mg is 5.2 ng/ml. It exerts its antitussive effect in 15-30 minutes after oral administration and the duration of action is approximately 5-6 hours.

Dextromethorphan is distributed to CNS, reaching detectable concentrations on cephalo-rachidien liquid (CRL). Studies in animals have shown cerebrospinal fluid/plasma concentrations ranging between 32.8 and 80%.

The drug undergoes a rapid and wide hepatic metabolism, originating an active metabolite, (dextrorphan) and two inactive (3-metoximorphinam and morphinam-3-1). Dextromethorphan is metabolised by the CYP2D6 isoenzyme, and undergoes oxidative demethylation.

Dextromethorphan and its demethylated metabolites are excreted primarily in urine. Elimination half-life of dextromethorphan is 1.4 to 3.9 hours and of dextrorphan of 3.4 to 5.6 hours.

5.3. Preclinical safety data

Dextromethorphan is a currently used medicine, included in practically all pharmacopoeias.

6. Pharmaceutical particulars**6.1. List of excipients**

Sucrose, anhydrous citric acid, methyl-paraben; liquid orange flavour; ethanol 96%, purified water.

6.2. Incompatibilities

In absence of incompatibilities this medicinal product must not be mixed with other medicinal products.

6.3. Shelf-life

3 years.

6.4. Special precautions for storage

No special precautions for storage are needed.

6.5. Nature and contents of container

Carton packs containing a glass bottle, type III, amber colour, and a leaflet. The bottle contains 200 ml-syrup. Each package contains a dosing cup.

6.6. Instructions for use and handling

No special requirements.

7. Marketing Authorisation Holder

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À Avenida da Siderurgia Nacional
4745-457 S. Mamede do Coronado
Portugal

8. Marketing Authorisation Number(s)

9. Date of first Authorisation

10. Date of revision of the text